Rakuten Product Classification — Visualizations 3+

This notebook contains the **visualizations from the newer audit notebook starting with Figure 3**, but adapted so it can run inside the existing `liora-capstone` GitHub project.

The data is loaded at the beginning using the project's existing helper:

```
df = load_merged_data()
```

Figures 1 and 2 from the newer notebook are intentionally omitted.

1. Imports

```
Environment loaded successfully.
```

2. Load Data

```
Dataset shape: (84916, 5)
```

	designation	description	productid	imageid	prdtypecode
0	Olivia: Personalisiertes Notizbuch / 150 Seiten / Punktraster / Ca Din A5 / Rosen-Design	NaN	3804725264	1263597046	10
1	Journal Des Arts (Le) N° 133 Du 28/09/2001 - L'art Et Son Marche Salon D'art Asiatique A Paris - Jacques Barrere - F...	NaN	436067568	1008141237	2280
2	Grand Stylet Ergonomique Bleu Gamepad Nintendo Wii U - Speedlink Pilot Style	PILOT STYLE Touch Pen de marque Speedlink est 1 stylet ergonomique pour GamePad Nintendo Wii U. Pour un confort ...	201115110	938777978	50
3	Peluche Donald - Europe - Disneyland 2000 (Marionnette À Doigt)	NaN	50418756	457047496	1280
4	La Guerre Des Tuques	Luc a des idées de grandeur. Il veut organiser un jeu de guerre de boules de neige et s'arranger pour en &eci...	278535884	1077757786	2705

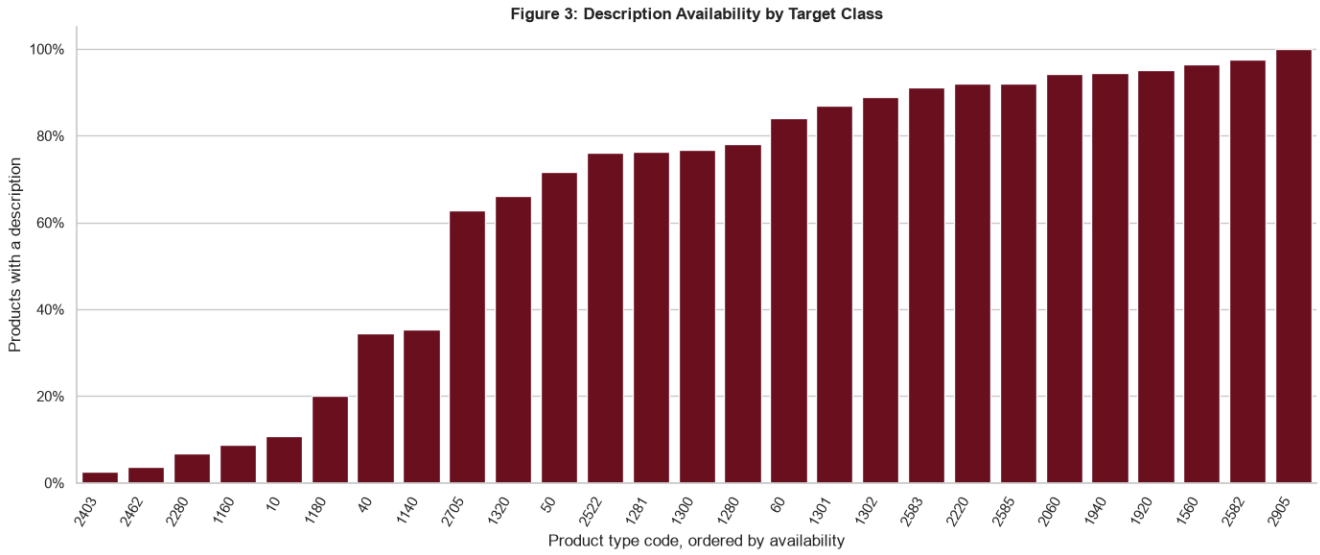
3. Audit Feature Construction

The following analytical features are required by Figures 3 onward.
The original dataframe `df` remains unchanged.

	designation	has_description	designation_chars	description_chars	total_text_chars
0	Olivia: Personalisiertes Notizbuch / 150 Seiten / Punktraster / Ca Din A5 / Rosen-Design	False	88	0	88
1	Journal Des Arts (Le) N° 133 Du 28/09/2001 - L'art Et Son Marche Salon D'art Asiatique A Paris - Jacques Barrere - F...	False	206	0	206
2	Grand Stylet Ergonomique Bleu Gamepad Nintendo Wii U - Speedlink Pilot Style	True	76	760	836
3	Peluche Donald - Europe - Disneyland 2000 (Marionnette À Doigt)	False	63	0	63
4	La Guerre Des Tuques	True	20	213	233

4. Figure 3 — Description Availability by Class

If missing descriptions are associated with the target, description availability itself may contain category-related information.



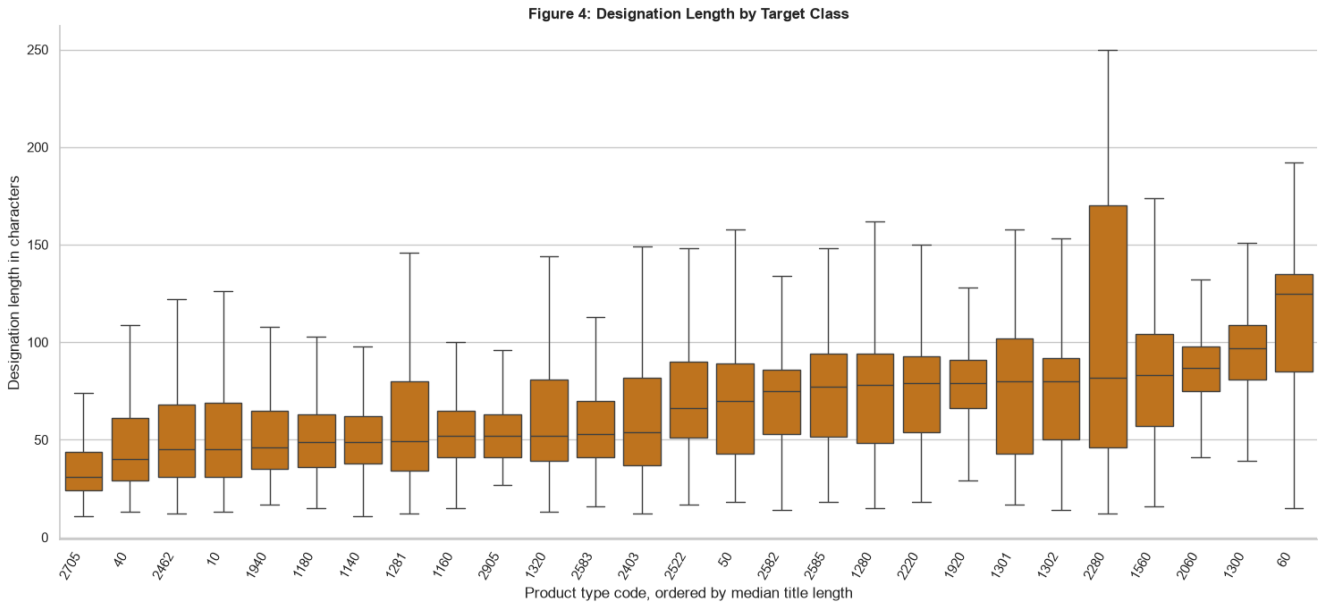
Finding 3 — missing descriptions are category-dependent

Observation. Class-level description availability ranges from 2.6% to 100.0%.

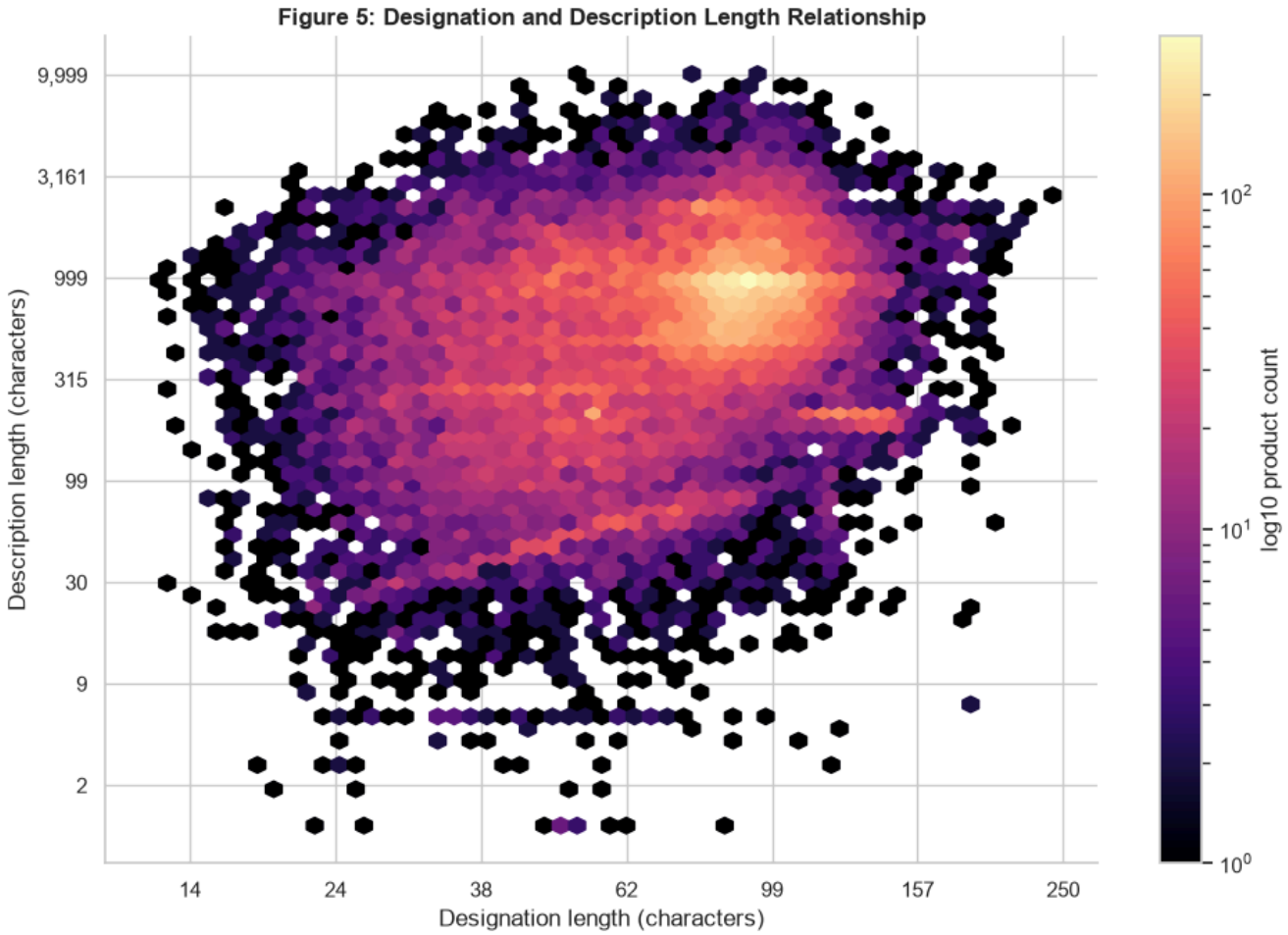
Quantitative validation. χ^2=43,757.8, p=0, Cramér's V=0.718.

Business implication. Description presence may improve classification, but it can also reflect seller behaviour rather than product semantics.

5. Figure 4 — Designation Length by Class

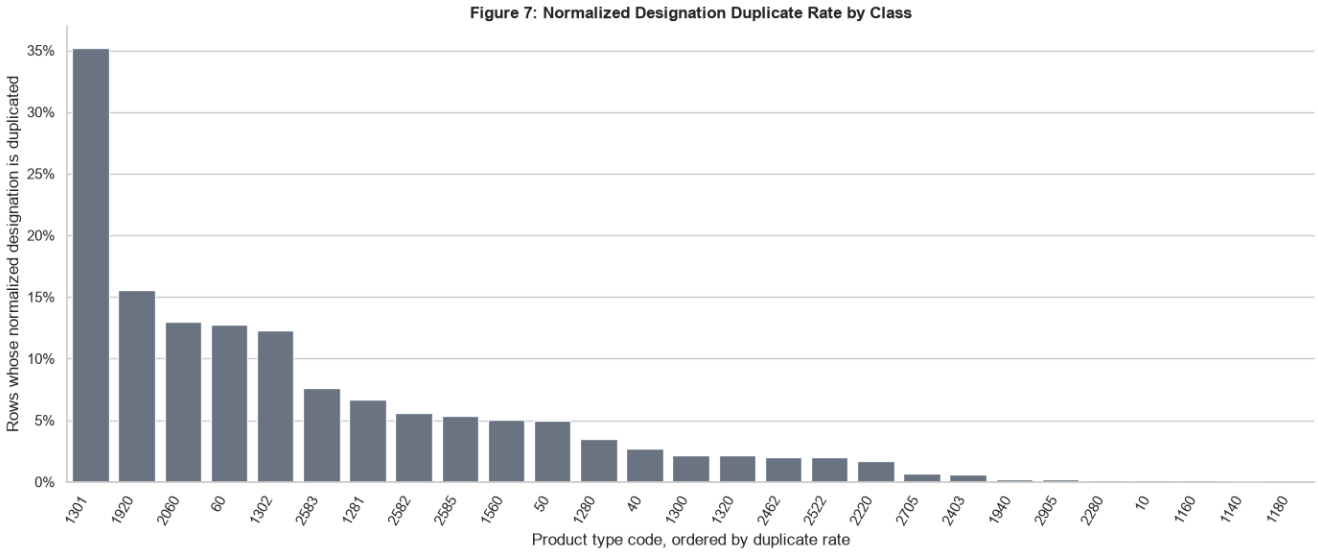


6. Figure 5 — Relationship Between Designation and Description Length



“Descriptions are much longer than designations, while their lengths are only moderately correlated. Therefore, we should test whether combining both text fields improves classification.”

8. Figure 7 — Duplicate Designations and Conflicting Labels



Finding 7 — duplicates create a validation-leakage risk

Observation. 4,172 training rows (4.9%) belong to a repeated normalized designation. 497 rows belong to duplicate-title groups containing more than one target class.

Quantitative validation. There are 1,463 repeated-title groups and 160 conflicting-label groups.

Business implication. Compare a standard stratified split with a grouped split based on normalized designation. Review conflicting groups before modeling.

4.172 Zeilen = 4,9 % des gesamten Trainingsdatensatzes haben einen normalisierten Titel, der mindestens zweimal vorkommt. Von diesen Duplikaten gehören 497 Zeilen zu problematischen Gruppen, bei denen derselbe Titel unterschiedliche Klassenlabels besitzt.

Titel Klasse

"Samsung Galaxy S20" 10 "Samsung Galaxy S20" 10 ← normales Duplikat

"Harry Potter DVD" 40 "Harry Potter DVD" 50 ← conflicting duplicate

A **duplicate group** contains all rows that have the same normalized designation.

Example:

Row	Normalized designation	Class
1	iphone 12 black	10
2	iphone 12 black	10
3	iphone 12 black	10
4	harry potter dvd	40
5	harry potter dvd	50
6	sony headphones	30
7	sony headphones	30

This gives:

- **7 duplicate rows**
- **3 duplicate groups**
  - `iphone 12 black`
  - `harry potter dvd`
  - `sony headphones`

Only one group is conflicting:

- `harry potter dvd` → appears with class 40 and class 50

So:

- Duplicate rows = 7
- Duplicate groups = 3
- Conflicting groups = 1

9. Text-Quality Flags

	train
contains_html	18.42%
contains_url	0.24%
contains_mojibake	1.08%

	flag	train_rows
0	contains_html	15645
1	contains_url	203
2	contains_mojibake	918

10. Figure 8 — Image Readability and Basic Properties

A random sample of training images is checked for readability and basic properties such as image dimensions, aspect ratio, and color mode. This helps identify missing or corrupted images before model training.

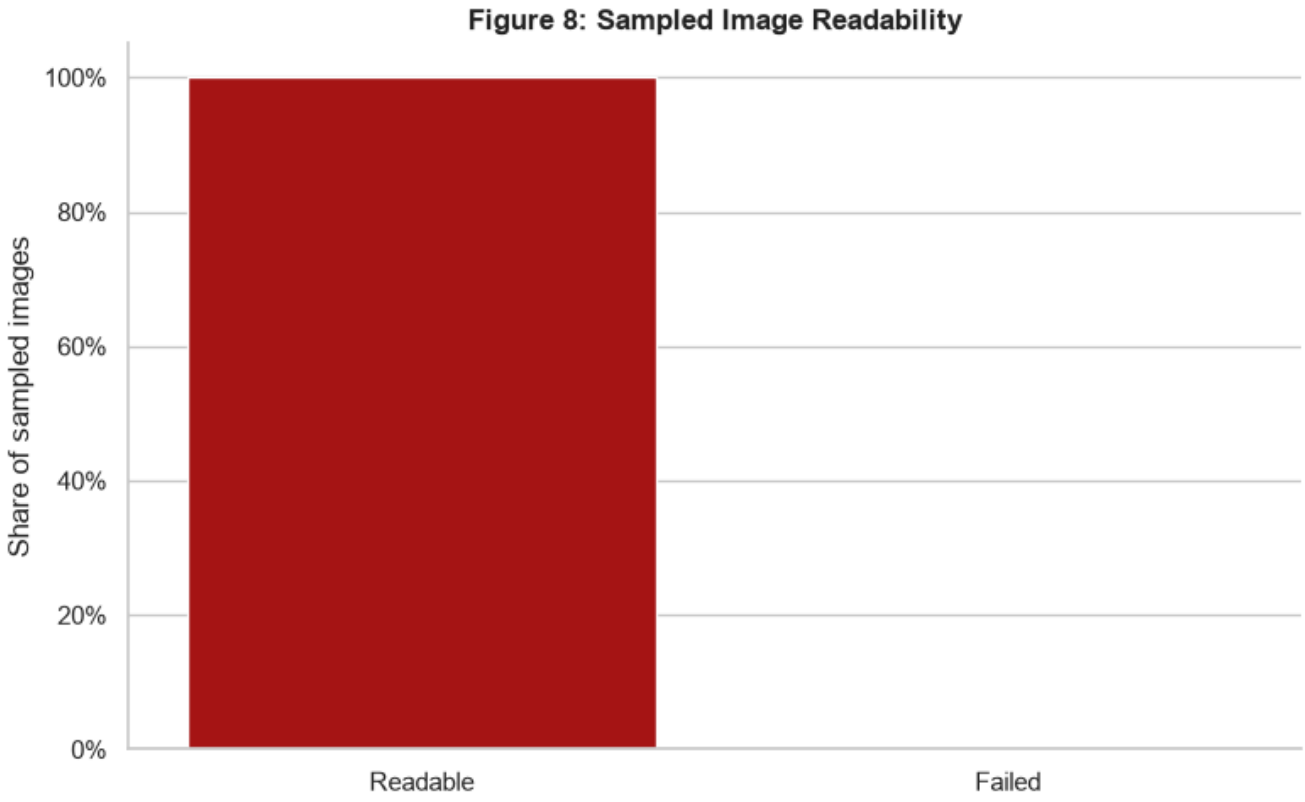
	width	height	aspect_ratio	mode	readable
0	500	500	1.0	RGB	True
1	500	500	1.0	RGB	True
2	500	500	1.0	RGB	True
3	500	500	1.0	RGB	True
4	500	500	1.0	RGB	True

```
Readable sampled images: 2,000/2,000
```

```
/var/folders/rq/rykc9bnx5nb4ypkbxvmrcjp80000gn/T/ipykernel_89556/1959709386.py:3: F
utureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.1
4.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  ax = sns.barplot(
```



	width	height	aspect_ratio
median	500.0	500.0	1.0
min	500.0	500.0	1.0
max	500.0	500.0	1.0

11. Preprocessing Decisions Supported by Figures 3+

- Keep `has_description` as an explicit feature if validation confirms that it helps.
- Compare designation-only, description-only and combined-text models.
- Use word and character representations for text.
- Be careful with duplicated normalized designations when constructing a validation split.
- Keep image loading centralized through `src.data_utils`.
- Figure 6 requires a project-level test-data loader; all other sections run using the existing `load_merged_data()` workflow.